Summary Statistics

Adverse Events Data: 1000-record Sample Dataset

Table 1: Adverse Events by Manufacturer

Date Range (MFG Date Received): January 11, 2022 – August 11, 2025

OMI (Orca1 Mfg. Index)	Death	Injury	Malfunction	TOTAL
ABBOTT LABORATORIES	5	61	57	123
BECTON, DICKINSON & CO.	0	10	83	93
Boston Scientific Corporation	19	152	247	418
Medtronic, Inc.	13	46	79	138
Teleflex	2	19	207	228
TOTAL	39	288	673	1000

Table 2: Top 20 Device Problem Codes by Manufacturer
Date Range (MFG Date Received): January 11, 2022 – August 11, 2025

Maude Device Problem	ABBOTT LABORATORIES	BECTON, DICKINSON & CO.	Boston Scientific Corporation	Medtronic, Inc.	Teleflex	Total
Material Rupture (1546)	10	47	177	0	1	235
Adverse Event Without Identified Device or Use Problem (2993)	21	2	148	47	2	220
Break (1069)	2	12	13	19	9	55
Material Twisted/Bent (2981)	6	2	2	0	42	52
Material Deformation (2976)	4	11	13	10	12	50
Deformation Due to Compressive Stress (2889)	0	0	0	8	38	46
Difficult to Remove (1528)	5	22	9	5	5	46
Leak/Splash (1354)	10	7	6	8	12	43
Detachment of Device or Device Component (2907)	0	18	11	13	1	43
Air/Gas in Device (4062)	30	0	10	0	2	42
Material Separation (1562)	3	0	1	0	37	41
Difficult to Advance (2920)	10	0	19	8	2	39
Physical Resistance/Sticking (4012)	0	0	1	11	20	32
Material Split, Cut or Torn (4008)	21	2	0	0	6	29
Material Integrity Problem (2978)	0	0	9	16	1	26
Entrapment of Device (1212)	0	5	11	2	2	20
Deflation Problem (1149)	2	14	2	1	0	19
Inflation Problem (1310)	2	9	7	0	0	18
Unraveled Material (1664)	0	5	0	0	13	18
Fluid/Blood Leak (1250)	0	4	0	0	10	14

Table 3: Device Problem Codes by Manufacturer associated with Air Embolism
Date Range (MFG Date Received): January 11, 2022 – August 11, 2025

Maude Device Problem	ABBOTT LABORATORIES	Boston Scientific Corporation	Medtronic, Inc.	Teleflex	Total
Adverse Event Without Identified Device or Use Problem (2993)	5	61	1	1	68
Air/Gas in Device (4062)	28	9	0	1	38
Material Integrity Problem (2978)	0	5	0	0	5
Leak/Splash (1354)	1	1	0	0	2
Insufficient Information (3190)	0	0	0	2	2
Improper or Incorrect Procedure or Method (2017)	2	0	0	0	2
Difficult to Flush (1251)	0	1	0	0	1
Difficult to Insert (1316)	1	0	0	0	1
Fluid/Blood Leak (1250)	0	0	0	1	1
Gas/Air Leak (2946)	0	1	0	0	1
Difficult to Advance (2920)	1	0	0	0	1
Inflation Problem (1310)	0	1	0	0	1
Device-Device Incompatibility (2919)	0	1	0	0	1
Defective Device (2588)	0	1	0	0	1
Material Deformation (2976)	0	1	0	0	1
Appropriate Term/Code Not Available (3191)	0	0	1	0	1
Material Split, Cut or Torn (4008)	1	0	0	0	1
Material Twisted/Bent (2981)	1	0	0	0	1
Mechanical Problem (1384)	0	0	1	0	1
Use of Device Problem (1670)	1	0	0	0	1
Total	41	82	3	5	131

Table 4: Top 20 Patient Problem Codes by Manufacturer

Date Range (MFG Date Received): January 11, 2022 – August 11, 2025

Patient_Codes	ABBOTT LABORATORIES	BECTON, DICKINSON & CO.	Boston Scientific Corporation	Medtronic, Inc.	Teleflex	Total
No Clinical Signs, Symptoms or Conditions (4582)	61	84	253	80	174	652
Air Embolism (1697)	33	0	75	3	5	116
Non specific EKG/ECG Changes (1817)	22	0	41	2	0	65
Thrombosis/Thrombus (4440)	4	1	39	3	1	48
Insufficient Information (4580)	1	0	0	12	30	43
Low Blood Pressure/ Hypotension (1914)	10	0	25	1	3	39
Pericardial Effusion (3271)	6	0	29	3	0	38
Hematoma (1884)	7	0	2	7	3	19
Cardiac Arrest (1762)	4	0	13	0	2	19
Hemorrhage/Blood Loss/Bleeding (1888)	9	0	3	4	2	18
Foreign Body In Patient (2687)	1	1	6	1	7	16
Stroke/CVA (1770)	2	0	12	2	0	16
Obstruction/Occlusion (2422)	1	1	2	11	0	15
Cardiac Perforation (2513)	1	0	13	0	0	14
Cardiac Tamponade (2226)	1	0	9	3	0	13
Bradycardia (1751)	3	0	8	0	1	12
Arrhythmia (1721)	5	0	7	0	0	12
Vascular Dissection (3160)	4	0	2	4	0	10
Perforation of Vessels (2135)	2	1	2	4	0	9
Embolism/Embolus (4438)	0	0	2	7	0	9

Table 5: Patient Problem Codes co-occurring with Air Embolism

Date Range (MFG Date Received): January 11, 2022 – August 11, 2025

# of Co-occurring codes	Co-occurring Codes	Count of Records	% of Air Embolism Records
0	Air Embolism only	26	22.4%
1	• Non specific EKG/ECG Changes (1817) • Arrhythmia (1721) • Low Blood Pressure/ Hypotension (1914) • Stroke/CVA (1770) • Cardiac Arrest (1762) • Transient Ischemic Attack (2109) • Convulsion/Seizure (4406) • Brain Injury (2219) • Nervous System Injury (2689) • Tachycardia (2095) • Pericardial Effusion (3271) • Muscle Weakness/Atrophy (1967) • Bradycardia (1751)	43	37.1%
2	• Non specific EKG/ECG Changes (1817), Ventricular Fibrillation (2130) • Low Blood Pressure/ Hypotension (1914), Non specific EKG/ECG Changes (1817) • Heart Block (4444), Non specific EKG/ECG Changes (1817) • Non specific EKG/ECG Changes (1817), Stroke/CVA (1770) • Cardiac Arrest (1762), Non specific EKG/ECG Changes (1817) • Loss of consciousness (2418), Numbness (2415) • Cognitive Changes (2551), Stroke/CVA (1770)	21	18.1%

# of Co-occurring codes	Co-occurring Codes	Count of Records	% of Air Embolism Records
	• Bradycardia (1751), Cardiac Arrest (1762) • Arrhythmia (1721), Non specific EKG/ECG Changes (1817) • Bradycardia (1751), Non specific EKG/ECG Changes (1817) • Non specific EKG/ECG Changes (1817), Tachycardia (2095) • Cardiac Arrest (1762), Low Blood Pressure/ Hypotension (1914)		
3	• Bradycardia (1751), Low Blood Pressure/ Hypotension (1914), Non specific EKG/ECG Changes (1817) • Atrial Fibrillation (1729), Heart Block (4444), Non specific EKG/ECG Changes (1817) • Asystole (4442), Non specific EKG/ECG Changes (1817), Vomiting (2144) • Angina (1710), Atrial Fibrillation (1729), Hypoxia (1918) • Convulsion/Seizure (4406), Non specific EKG/ECG Changes (1817), Stroke/CVA (1770) • Arrhythmia (1721), Cardiac Arrest (1762), Low Blood Pressure/ Hypotension (1914) • Cardiac Arrest (1762), Diminished Pulse Pressure (2606), Movement Disorder (4412) • Cardiac Arrest (1762), Low Blood Pressure/ Hypotension (1914), Stroke/CVA (1770) • Arrhythmia (1721), Hemorrhage/Blood Loss/Bleeding (1888), Low Blood Pressure/ Hypotension (1914) • Decreased Level of Consciousness (4591), Low Blood Pressure/ Hypotension (1914), Stroke/CVA (1770) • Low Blood Pressure/ Hypotension (1914), Non specific EKG/ECG Changes (1817), Stroke/CVA (1770) • Bradycardia (1751), Loss of consciousness (2418), Stroke/CVA (1770) • Arrhythmia (1721), Low Blood Pressure/ Hypotension (1914), Non specific EKG/ECG Changes (1817)	15	12.9%
4	• Dysphasia (2195), Nervous System Injury (2689), Paralysis (1997), Reduced Blood Flow (4598)	5	4.3%

# of Co-occurring codes	Co-occurring Codes	Count of Records	% of Air Embolism Records
	• Bradycardia (1751), Cardiac Arrest (1762), Loss of consciousness (2418), Low Blood Pressure/ Hypotension (1914) • Bradycardia (1751), High Blood Pressure/ Hypertension (1908), Non specific EKG/ECG Changes (1817), Stroke/CVA (1770) • Arrhythmia (1721), Low Blood Pressure/ Hypotension (1914), Non specific EKG/ECG Changes (1817), Stroke/CVA (1770) • Diminished Pulse Pressure (2606), Heart Block (4444), Hypoxia (1918), Non specific EKG/ECG Changes (1817)		
5	• Bradycardia (1751), High Blood Pressure/ Hypertension (1908), Low Blood Pressure/ Hypotension (1914), Non specific EKG/ECG Changes (1817), Tachycardia (2095) • Bradycardia (1751), Cardiac Arrest (1762), Low Blood Pressure/ Hypotension (1914), Non specific EKG/ECG Changes (1817), Stroke/CVA (1770) • Arrhythmia (1721), Asystole (4442), Hypoxia (1918), Low Blood Pressure/ Hypotension (1914), Respiratory Failure (2484)	3	2.5%
6	• Asystole (4442), Bradycardia (1751), Coma (2417), Dyspnea (1816), Low Blood Pressure/ Hypotension (1914), Non specific EKG/ECG Changes (1817) • Cardiac Arrest (1762), Low Blood Pressure/ Hypotension (1914), Non specific EKG/ECG Changes (1817), Obstruction/Occlusion (2422), Tachycardia (2095), Thrombosis/Thrombus (4440) • Brain Injury (2219), Heart Block (4444), Low Blood Pressure/ Hypotension (1914), Non specific EKG/ECG Changes (1817), Shock (2072), Ventricular Fibrillation (2130)	3	2.5%

Table 6: Example Themes

Theme Name	Key Narrative Cues	Linked Injury Type(s)	Representative Excerpts	Instructor Notes
Valve Open or Vacuum During Device Exchange	"valve open," "vacuum created," "air entered sheath," "exchange"	Air Embolism → Stroke / Hypoxia	"When the dilator was removed, the valve was noted to be open and air entered the system."	Strong support for H1 (procedural mechanism). Consistent with Field Safety Notice describing negative pressure and air ingress.
Deep Sedation or Non-Intubated Patient Context	"deep sedation," "no positive pressure," "patient not intubated," "spontaneously breathing"	Hypoxia, Stroke	"Procedure performed under deep sedation without intubation. Sudden desaturation and neurologic deficit observed."	Partial support for H2. Sedation context rarely explicit, but narratives imply absent airway control.
Device Exchange / Flushing Without Valve Closure	"catheter flushed before closing valve," "air observed in tubing," "bubbles seen"	Air Embolism (no injury or mild hypoxia)	"Technician flushed the device prior to closing valve, resulting in air introduction."	Common across manufacturers — procedural training gap.
Cross-Manufacturer Terminology Consistency	Identical phrasing across Abbott and Boston Sci: "air/gas in device," "device difficult to remove," "catheter rupture"	Air Embolism, Thrombus	"Report text mirrors other manufacturers' phrasing — suggests standardized report templates or shared mechanism."	Supports H3. Pattern similarity is qualitative, not numerical.
Incomplete or Ambiguous Narratives	"no clinical signs," "unable to determine cause," "insufficient information"	None / Unclear	"No patient injury was reported; cause could not be determined."	Neutral theme — teaches that not all narratives are informative. Reinforces need for structured + unstructured analysis.

Table 7: Hypothesis Testing

Hypothesis ID	Evidence from Narratives	Preliminary Conclusion
H1 – Procedural Mechanism	Narratives repeatedly describe **air ingress during sheath exchange** or **valve left open**, often with "vacuum," "air bubbles," or "valve open" phrasing. Serious injuries (stroke, hypoxia) co-occur. Language mirrors recall rationale.	**Supported.** Strong alignment between narrative patterns and Field Safety Notice on negative pressure and valve handling.
H2 – Clinical Setting	Only ≈ 10–15 % of narratives explicitly mention sedation; however, several describe procedures "under deep sedation" or "without intubation," and those events disproportionately include air embolism and hypoxia.	**Partially Supported.** Evidence suggests higher risk under deep/conscious sedation, but data insufficient for confirmation due to missing sedation fields.
H3 – Cross-Manufacturer Pattern	Similar terminology appears in Abbott and Boston Scientific reports: "air/gas in device," "material rupture," "difficult to remove." Both share top device and patient problem codes (Air Emb., Thrombosis, Material Rupture). Proportional scaling differs but relative ranking identical.	**Supported.** Cross-manufacturer phrasing and code patterns confirm shared procedural mechanisms rather than design defects.

Table 8: Thematic Map

Failure Mode / Procedural Factor	Injury / Harm Outcome	Manufacturer Response or Recall Theme	Instructor Notes
Valve Open During Sheath Exchange	Air Embolism → Stroke / Hypoxia	Field Safety Notice (2025) – Emphasize keeping valve closed and maintaining positive pressure	Core procedural mechanism behind recall; validates H1
Device Exchange or Flushing Without Valve Closure	Air Embolism (no injury or mild hypoxia)	User training advisory on flush/aspiration steps	Appears in Abbott and Boston Sci reports; reinforces H3
Deep Sedation / No Positive Pressure Ventilation	Hypoxia / Neurologic deficit / Stroke	Procedural recommendation – use positive pressure or intubation when appropriate	Partial support for H2; highlights missing structured data
Material Rupture or Difficult to Remove Catheter	Thrombosis / Vascular trauma (secondary)	Device handling and inspection guidelines	Shared across manufacturers; supports H3 procedural similarity
Incomplete or Ambiguous Narratives (“No Clinical Signs”)	None / Unknown harm potential undetected	Internal data quality improvement and post-market surveillance training	Teaching example of MAUDE under-reporting limitations